A Dormant Subspace, Not a Missing One: A Decodable but Causally Inert Answer-Site Representation of Arithmetic in Llama-3.1-8B

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Abstract

Large language models often fail at arithmetic composition even when they solve every part in isolation. We study a token-matched instance in Llama-3.1-8B: wrapping a multiplicative operand in an additive identity, $(\,0 + B\,) * C$, collapses multiplication accuracy (instruct 0.27, base 0.51) although the model evaluates the wrapped sub-expression perfectly (1.00) and multiplies inside a bracket (0.91). The failure is operation-specific (the additive analogue composes at 0.99), depth-sensitive, magnitude-controlled, and recovers with a few in-context examples. We then ask what the internal representation looks like. At the answer position the operand B and the near-product $B \cdot C$ are linearly decodable ($R^2 \approx 0.96$), which invites a "the value is there but unused" story. We show this reading is an interpretability illusion. The decode is operand-dominated: it has no margin over a linear-(B, C) baseline where there is no headroom, and only a small margin (gap ≈ 0.06, CI excludes zero) where there is. More decisively, the decodable direction is causally inert: steering along it does not change the emitted answer, and a full-residual swap at the answer position does not flip it either, while patching the operand positions does move it (recovery 0.50), a dead site, not a dead probe. Decodability at the answer site is not evidence of computation there. The collapse replicates across model families in matched format, with a prompt-format confound we make explicit.

1 Introduction

When does a decodable representation license a claim about computation? Probing the residual stream is a standard route to mechanistic claims: if an intermediate value decodes, the model is said to "know" it, and when a downstream failure co-occurs with a present-but-decodable value, that value is called computed-but-unused. We show this inference is unsafe, using arithmetic composition as a controlled testbed. A value that decodes at the answer position can be an operand-reconstructible, causally inert shadow of the computation, not the computation itself.

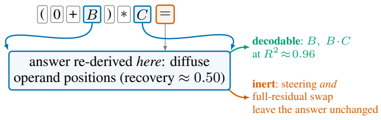

Figure 1: The answer-site subspace is *dormant*, not missing. At the "$=$" position the operand and near-product are linearly decodable, but that direction is causally inert (steering and a full-residual swap change nothing) and operand-explained; the answer is instead re-derived diffusely from the operand positions. Decodability at the answer site is not evidence of computation there.

Our vehicle is a single controlled failure. For $B, C \in [20, 49]$, wrapping the first operand in an additive identity, $(\,0{+}B\,){*}C\ {=}$, collapses accuracy to 0.27 (instruct) / 0.51 (base) although the model returns B for $(\,0 + B\,) =$ at 1.00 and multiplies inside a bracket, $(\,B * C\,)\ =$, at 0.91; the gap is large and tight ($\Delta(\text{C4}{-}\text{C1}) = 0.64$, 95% CI $[0.60, 0.69]$, instruct). This is a clean hook, not our contribution: we use it to ask whether a decodable representation at the answer site licenses a claim about computation. Both B and the product $B \cdot C$ are linearly decodable there ($R^2 \approx 0.96$), inviting the reading that the product "is represented but not used." Figure 1 summarizes why that reading fails.

Contributions.

- A token-matched, control-rich instance of a compositional arithmetic blind spot in Llama-3.1-8B: operation-specific, depth-sensitive, magnitude-controlled, and few-shot-recoverable, each with a confidence interval (§4).
- Evidence that the decodable answer-site subspace is *operand-dominated*: it has no margin over a linear-(B, C) baseline where the band

leaves no headroom, and only a small margin (gap ≈ 0.06, CI excludes zero) where it does, robust to a C1-independent probe-layer choice (§5).

- A causal test showing the decodable direction is *inert*: probe-direction steering and a full-residual swap at the answer position both leave the emitted answer unchanged, anchored by a positive control (patching the operand positions moves the answer, recovery 0.50), so the null is a dead site, not a dead probe. Ablating the decode direction preserves $\approx 85\%$ of the swap's (sub-flip) answer-effect (§5).
- A cross-family replication in matched bare format, together with an explicit prompt-format confound that we argue precludes reading the cross-model results as a clean family dissociation (§6).

We read the answer-site subspace as *dormant* in the sense of Makelov et al. (2024): present to a probe, invisible to the computation.

2 Related Work

Probing validity and causal probing. A linear probe measures correlation, not use: a probe can fit a random control task (Hewitt and Liang, 2019), and decodability alone licenses no functional claim (Belinkov, 2022). The information-theoretic view sharpens this: probing recovers information already present in the inputs (Pimentel et al., 2020), and description length, not accuracy, measures how *extractable* it is (Voita and Titov, 2020). This is exactly the objection our selectivity test answers: $B{\cdot}C$ is nearly linear in (B, C) on-distribution, so a probe recovers it whether or not the product is represented; the linear-(B, C) baseline quantifies that inherited decodability and the gap over it is what remains (§5, Prop. 1). Causal probing goes further by intervening: amnesic probing removes a property and reads the behavioral effect (Elazar et al., 2021), and concept-erasure methods formalize removing a linear subspace (?Belrose et al., 2023); our orthogonal-complement (perp) intervention is in that lineage, but applied as a single ridge direction inside a full-residual donor swap at one site, scored by a full-product metric over a dose grid. The failure mode we document is the interpretability illusion of Bolukbasi et al. (2021) and, for subspace activation patching specifically, Makelov et al. (2024): a direction can carry a decodable signal yet be causally disconnected from the output.

We adopt activation-patching best practice throughout (Zhang and Nanda, 2024; Heimersheim and Nanda, 2024) and read our interchange intervention through causal abstraction (Geiger et al., 2021) (§5).

Arithmetic and number representations. How language models compute arithmetic is increasingly well mapped. Numbers are encoded as structured directions (Wallace et al., 2019); addition is implemented with Fourier features (Zhou et al., 2024) and trigonometric identities (Kantamneni and Tegmark, 2025), and small models grok modular addition into a clean circuit (Nanda et al., 2023; Quirke and Barez, 2024). In large models Stolfo et al. (2023) trace arithmetic from operand tokens to the final position through late modules, and Nikankin et al. (2025) argue the computation is a diffuse "bag of heuristics" rather than an algorithm. Our result adds a representational corollary to the bag-of-heuristics picture: the decodable near-answer at the "=" position is an operand echo, not a stored intermediate; the answer is assembled from the operand positions at readout, so a value that decodes at "=" need not be computed there.

Elicitation and format sensitivity. Model accuracy is sensitive to surface form that leaves the answer invariant (Sclar et al., 2024), and mathematical errors in particular track surface structure and recover under rephrasing (Williamson et al., 2025; Reusch and Lehner, 2022). That our zero-shot collapse recovers with a few in-context examples is consistent with demonstrations acting largely through format and priors rather than content (Min et al., 2022); we treat the recovery as a control on the behavioral phenomenon, not as our contribution, and locate the mechanism question in the zero-shot regime. Closest to us, Matsumoto et al. (2022) find decoded intermediate values in neural math solvers to be causally usable; we find a decodable answer-site subspace that is causally inert, so whether a decodable value is used is a per-site empirical question.

3 Setup and Research Questions

Model and stimuli. We study Llama-3.1-8B (Grattafiori et al., 2024), base and instruction-tuned, in bare continuation format, by greedy decoding and integer parsing. Stimuli are $N = 400$ operand pairs (B, C) drawn once with a fixed seed from $[20, 49]$ and held constant across every condition,

	surface	base	instruct
C0	`B * C =`	0.838	0.848
C4	`( B * C ) =`	0.890	0.903
C6	`( 0 + B ) =`	1.000	1.000
C1	`( 0 + B ) * C =`	**0.508**	**0.265**
A1	`( 0 + B ) + C =`	n/a	0.995
D1	`(( 0 + B )) * C =`	n/a	0.038

Table 1: The composition blind spot (exact-match accuracy, N=400). The parts succeed (C4, C6) but the composition collapses (C1); Δ(C4−C1) = 0.64, CI [.60, .69] (instruct). A1/D1 are instruct-only. Full battery with Wilson intervals in Appendix A.

token-length matched, with spacing controlled. The central contrast is C1 $=$ $(\,0 + B\,) * C$ $=$ (the composed failure) against C4 $= (\,B * C\,) =$ (multiply inside a bracket, succeeds) and C6 $=$ $(\,0 + B\,) =$ (evaluate the wrapper, succeeds).

Metrics. Accuracies carry Wilson intervals; paired contrasts carry paired bootstrap intervals (10,000 resamples). Decodability is five-fold cross-validated ridge R^2 (item-bootstrap interval; selectivity-gap CIs use 300 resamples given the per-draw cross-validation cost) at a named token position. Selectivity compares the decode against a linear-(B, C) baseline (a probe given only the raw operands and Gaussian noise), so a high R^2 that is merely linear in the operands does not count as representing the product. Causal tests use symmetric operand counterfactuals and a full-answer log-probability metric $\Delta = \log p(P') - \log p(\text{true})$; because Llama chunks the leading digits, a first-token score is non-discriminative between two products.

Research questions. **RQ1**: is the composition failure a genuine, controlled effect or an artifact of difficulty, magnitude, or elicitation (§4)? **RQ2**: is the near-product decodable at the answer site a *represented* quantity or reconstructible from the operands alone (§5)? **RQ3**: is the decodable direction causally used to produce the answer (§5)?

4 The Blind Spot

The model evaluates every ingredient of $(\,0 + B\,) * C$ yet fails to compose them (Table 1). We establish that this is a real effect, not an artifact, with four controls, and we state each at the precision the data support.

It is operation-specific, not format brittleness. Replacing the outer multiplication with addition, $(\,0 + B\,) + C\ =$, restores accuracy to 0.995 (instruct) against 0.265 under *identical* format; $\Delta(\text{A1} - \text{C1}) = 0.730$, 95% CI $[0.69, 0.77]$. It is also spacing-invariant: with all spaces removed the inside-bracket multiply still survives (C8 $=$ 0.785 instruct, 0.765 base) while the composition still collapses (C7 $=$ 0.19, 0.018), the same dissociation. So it is not generic spacing or parenthesis sensitivity; specifically the multiplicative composition breaks. (A1/A2 are instruct-only; we do not extrapolate to base.)

It is controlled, not an artifact. Three further controls hold. *Depth* is a distinct stressor: a redundant nesting $(\,(\,0 + B\,)\,) * C =$ drops to 0.038, but nesting degrades *addition* too (instruct nested addition falls to 0.25 from 0.99), so we treat depth as a general parsing stressor, not more composition evidence. *Magnitude* is ruled out: within every matched-$|B \cdot C|$ bin C1 stays well below C4 (0.53 vs 0.96 smallest, 0.00 vs 0.70 largest). And the collapse is an *elicitation* fragility: two to four fixed-seed in-context examples recover C1 to the C4 ceiling (instruct $0.265 \to 0.885\,[.85, .91] \to 0.915\,[.88, .94]$; base $0.508 \to 0.838 \to 0.882$), so it is not an inability to multiply.

Two overstatements we avoid. The effect is neither universal (base $(\,B + 0\,) * C$ barely moves, drop 0.015, vs 0.33 for $(\,0 + B\,)$; Appendix B) nor uniformly worse under tuning (on plain parentheses instruct is *better*, 0.755 vs 0.495); and C0, not 1.0, is the reference ceiling ($B * C =$ is 0.848 instruct).

5 A Dormant Subspace

The seductive story is that the product is computed and represented at the answer position but not read out. We test it in three steps and reject it.

The product decodes (RQ2, correlational). A cross-validated probe recovers $B \cdot C$ from the "=" residual at $R^2 = 0.965$ (instruct), 0.967 (base) and the operand B at 0.96–0.97, with B decodable at ")" at *every* layer including the last (instruct 0.98, layer 31); taken alone this looks like a represented product.

But the decode is operand-dominated (RQ2, selectivity). Decodability of $B{\cdot}C$ need not mean the product is represented, since $B \cdot C$ is itself largely linear in B, C over a narrow band. We test this with a selectivity gap over a linear-(B, C) baseline,

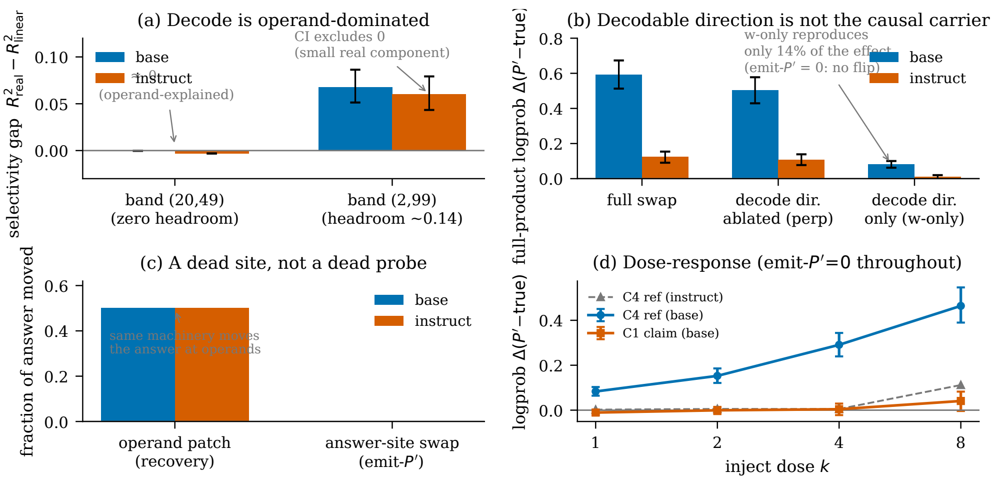

Figure 2: The decodable answer-site subspace is dormant (bars/error bars are 95% CIs). **(a)** Selectivity gap (decode R^2 for $B \cdot C$ at "=" minus a linear-(B, C) baseline): at $[20, 49]$ no headroom (gap ≈ 0, operand-explained); at $[2, 99]$ a small gap whose CI excludes zero. **(b)** Causal share of a full-residual swap at the C4 "=" (base): ablating the decode direction (perp) preserves most of the effect, the decode direction alone (w-only) reproduces little, CIs non-overlapping; the swap flips no answers (emit-$P' = 0$). **(c)** The dissociation as a fraction of the answer moved (two metrics): patching moves the answer at the operand positions (recovery 0.50) but the answer-site swap flips nothing, a dead *site*, not a dead probe. **(d)** Inject dose-response on the same axis: the C1 claim site is flat at every dose (CIs bracket 0) while the C4 reference rises monotonically (base) yet never flips; instruct's reference is weak and non-monotone.

but the gap test only has resolving power when the baseline leaves headroom:

Proposition 1 (Headroom bound)**.** *For a target g, a decode probe with cross-validated $R^2_{\mathrm{real}} \leq 1$, and a baseline feature class F with cross-validated R^2_F on distribution D, the selectivity gap obeys* gap $=$ $R^2_{\mathrm{real}} - R^2_F \leq 1 - R^2_F$. *The gap test therefore resolves at most the* headroom $1 - R^2_F$, *so headroom is a design statistic that must be reported.*

(Proof in Appendix F; it applies to any derived quantity nearly linear in its inputs on-distribution.) We compare $B \cdot C$ against the linear-(B, C) baseline (Figure 2a). At $[20, 49]$ the baseline reaches $R^2 = 0.968$: by Prop. 1 the headroom is only 0.032, so the test is a priori near-vacuous, and indeed the gap is -0.0002. At $[2, 99]$, where $B \cdot C$ is nonlinear in the operands (headroom 0.135), a small gap appears: $+0.068$, 95% CI $[0.051, 0.086]$ (base) and $+0.060$, $[0.043, 0.079]$ (instruct), both excluding zero, and still positive at a layer chosen by a C1-independent criterion (the C4 decode peak) rather than the C1 argmax. We flag that $[2, 99]$ is a harder regime where the parts fall below our 0.80 floor (C4 $= 0.71$ base, 0.73 instruct), so we read it as a stress test, not a comfortably in-scope measurement, and rest the headline on the band-independent causal null below. The decode is operand-dominated (≥ 0.94 of it is reproduced by the linear-(B, C) baseline, ≈ 1.0 at $[20, 49]$) with at most a small operand-nonlinear component, consistent with a weak product signal (behavior-tracking: C4 gap $>$ C1 gap, difference CI excludes zero).

And the decodable direction is causally inert (RQ3). We steer the "=" residual along the fitted direction to write a counterfactual product P', on held-out items. On the first-token metric the inject is null at every layer and matches norm-matched random and shuffled controls (Appendix D), but that metric is non-discriminative between products. On the discriminative full-product metric, injecting at the C1 "=" is flat at every dose $k \in \{1, 2, 4, 8\}$ (CIs bracket zero) while the same inject at the succeeding C4 "=" rises monotonically ($+0.08 \to +0.46$, every CI excluding zero, base) yet never flips: the axis has dynamic range where the product is used and none at the failing site, a dead direction rather than a dead instrument (emit-P' stays zero throughout, a saturating criterion, so we read the log-probability). A full-residual swap at the C4 "=", *where the model succeeds*, likewise fails to transplant the donor product (emit-$P' = 0$; log-probability $+0.59$ at L19 where we decompose it, peak $+0.66$ at L23, still sub-flip); after the swap the model still emits its *own* correct answer at 0.90–0.95 across L19–L31, so the site is demonstrably bypassed. The product is thus not linearly resident at "=" even where computed correctly, a fortiori at C1. The one intervention that moves the answer is at the operand positions (recovery 0.50; a clean-answer recovery metric, a different readout and operation).

Certifying the illusion. We decompose the C4 swap effect into the decode direction and its orthogonal complement (Figure 2b). Ablating the decode direction still leaves $+0.50$ (base), while the decode direction alone contributes only $+0.08$, with non-overlapping CIs: $\approx 85\%$ of the swap's (sub-flip) answer-effect survives removing the decodable direction (the swap moves only $\approx 20\%$ of its magnitude along it). (The instruct swap effect is smaller, $+0.12$, its decode-direction component not significantly positive; we report the decomposition for base.) We use *dormant* strongly: unlike a patch-potent-but-unused component, this direction moves essentially nothing even when patched, present to a probe but invisible to computation (Makelov et al., 2024). Consistent with Stolfo et al. (2023) and Nikankin et al. (2025), the operands carry the signal but no sparse head set reconstructs it (best head < 0.02); the value at "=" decodes and correlates yet cannot be steered.

As a causal-abstraction claim. The full residual swap is an interchange intervention (Geiger et al., 2021) on the "=" residual: substituting a donor's representation should, if $B \cdot C$ is computed and stored there, install the donor's answer. Interchange accuracy is 0 (emit-$P' = 0$) and the original answer survives (emit-true 0.90–0.95), so the hypothesis "$B \cdot C$ is computed at ("=", layer l)" fails as a causal-abstraction claim at every late l, even though the correlational decode holds. Decodability and causal realization come apart at this node.

Hypotheses and evidence. Table 2 collects the three readings of the failure against the committed evidence.

Predictions. H2 is falsifiable. It predicts that (i) ablating attention from "=" to the operand positions during generation should destroy C4 accuracy, and (ii) few-shot recovery should act at the operand positions rather than at "=". We state these as predictions, not results.

Hypothesis	Verdict	Evidence (committed)
H1: product *stored* at "="	rejected	swap emit-P'=0, dose C1 flat, emit-true 0.90–0.95
H2: *assembled* at readout from operands	supported	operand recovery 0.50; B decodable at every layer incl. 31; emit-true 0.90–0.95
H3: sparse answer-site circuit	no locus	best head $<$ 0.02; all heads inconclusive

Table 2: The composition failure adjudicated. Every cell traces to a committed file (Appendix A; `lateswap`, `dose_response`, `operand_localization`, `position_decodability`, `head_path_patch`).

6 Generality

In matched bare format the collapse is not specific to Llama-3.1-8B. Among models that clear a parts floor (C4, C6 ≥ 0.80), the composed form collapses for Gemma-2-9b-it (C1 $=$ 0.043, C4 $=$ 0.99) and Llama-3.2-3B (C1 $=$ 0.36), alongside base and instruct Llama-3.1-8B; three others fail the parts in bare format and are out of scope, not counterexamples (Appendix E). Chat formatting changes this, not cleanly: Gemma-2-9b-it *composes* in its native chat template (C1 $=$ 0.95), but there Llama-3.1-8B falls below the parts floor (C4 $=$ 0.70) and leaves scope, so the composing and failing models are never compared in the same format. We read this as replication in matched bare format plus a prompt-format confound, *not* as evidence that some families are immune; per-model native-format evaluation is the right axis for a clean cross-family claim, which we do not make.

7 Conclusion

A parenthesized additive identity selectively breaks multiplicative composition in Llama-3.1-8B, and the "represented but unused" reading of the decodable answer site is an interpretability illusion: the near-answer at "=" is operand-dominated and causally inert, an operand echo, while the answer is assembled from the operand positions at readout. The methodological corollary is a short protocol (Appendix G): a decodability claim about computation needs a selectivity margin over a stated baseline with headroom reported (Prop. 1), a causal test on a discriminative metric with a same-axis reference that moves, and a positive control at a site that moves the answer. A high decodability R^2 licenses none of this on its own.

Limitations

Our mechanistic claims are established on one model family (Llama-3.1-8B, base and instruct); the cross-model results (§6) are behavioral replication in matched format, not mechanistic transfer, and we deliberately make no cross-family dissociation claim. The causal localization is distributed: the operand positions carry the answer-relevant signal but no sparse head set reconstructs it, so we characterize *where* the computation lives, not a minimal circuit. The selectivity gap that excludes zero is small and appears only at the wider band $[2, 99]$, which is a harder regime where the parts themselves fall below our 0.80 floor (C4 $= 0.71/0.73$); we therefore read it as a stress test of operand-domination and rest the headline on the band-independent causal null, with the dormancy certified by the causal-share decomposition rather than the static gap. The operation and depth controls (A1, A2, D1) are committed for the instruct model only, and the C4 dose reference is base-scoped: on instruct the same-axis reference is weak (a single dose crosses, non-monotone), so the dynamic-range argument for instruct rests on the operand-route positive control rather than the answer-site dose curve. That positive control is a clean-answer recovery metric at the operand positions, a different readout and operation than the answer-site full-product swap; a same-metric answer-site positive control is the natural next experiment (its notebook is included in the released code), as is a second wrapper family beyond the additive identity. Finally, the linear-(B, C) baseline's span excludes B^2, C^2, and $B \cdot C$, so the gap upper-bounds operand-nonlinear structure and is consistent with, but does not by itself identify, a product component; a quadratic-feature baseline is the decisive follow-up.

Availability. Code, data, figure scripts, and every control (including the two next-experiment notebooks above) are released at an anonymized mirror, `https://anonymous.4open.science/r/operator-precedence-anon`, with each reported number mapped to its source file.

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A Full Battery and Confidence Intervals

Table 3 gives every condition with Wilson 95% intervals ($N = 400$). Base A1/A2/D1 are not in our committed data and are omitted rather than estimated.

	surface	base	instruct
C0	`B * C =`	.838 [.80,.87]	.848 [.81,.88]
C1	`( 0 + B ) * C =`	.508 [.46,.56]	.265 [.22,.31]
C2	`( 0 + B * C ) =`	n/a	.678 [.63,.72]
C3	`( B ) * C =`	n/a	.755 [.71,.80]
C4	`( B * C ) =`	.890 [.86,.92]	.908 [.88,.93]
C5	`0 + B * C =`	n/a	.542 [.49,.59]
C6	`( 0 + B ) =`	1.00 [.99,1.0]	1.00 [.99,1.0]
C7	`(0+B)*C=`	.018 [.01,.04]	.190 [.16,.23]
C8	`(B*C)=`	.765 [.72,.80]	.785 [.74,.82]
A1	`( 0 + B ) + C =`	n/a	.995 [.98,1.0]
A2	`0 + ( B + C ) =`	n/a	.928 [.90,.95]
D1	`(( 0 + B )) * C =`	n/a	.038 [.02,.06]

Table 3: Full battery, exact-match accuracy with Wilson 95% CIs. Note C5: relying on bare operator precedence (`0 + B * C =`, 0.542) beats the parenthesized additive identity (C1, 0.265), so the parenthesization actively hurts (instruct).

Magnitude control. Within matched $|B \cdot C|$ bins (instruct), C1 vs C4: $[440, 813)$ .53/.96 (n=79); $[813, 1185)$ .28/.94 (126); $[1185, 1558)$ .16/.89 (104); $[1558, 1930)$ .19/.89 (64); $[1930, 2303)$ .00/.70 (27). Every paired difference has a positive bootstrap CI.

B Wrapper Blind-Spot Map

C By-Layer Decodability

D Steering Nulls

E Cross-Model Battery

Matched bare format, in scope if C4 and C6 $\geq$ 0.80: Llama-3.1-8B base C1 .508 / instruct .265; Gemma-2-9b-it .043 (C4 .99); Llama-3.2-3B .358 (C4 .88). Out of scope (parts fail): Qwen2.5-7B (C4 .00), Mistral-7B (C4 .62), Llama-3.2-1B (C4 .31). Chat format: Gemma-2-9b-it composes (C1 .95, C4 .99) but Llama-3.1-8B chat falls out of

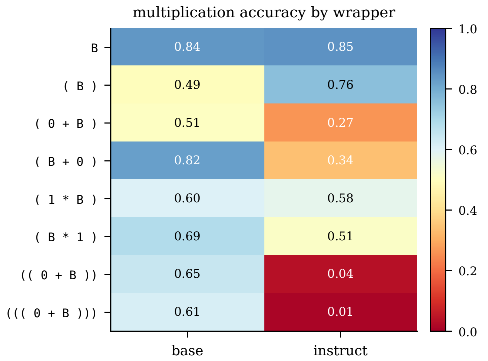

Figure 3: Multiplication accuracy by wrapper. Position-dependence is a base-model effect (left-placed $(\,0{+}B\,)$ collapses, right-placed $(\,B{+}0\,)$ barely moves; instruct degrades on both placements), and instruction tuning is not uniformly worse (plain parentheses).

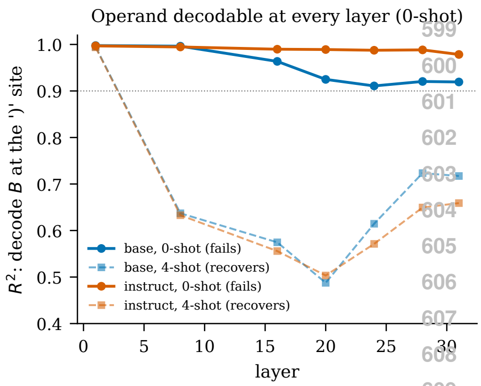

Figure 4: Operand B decodes at every layer in the failing zero-shot regime (solid). The few-shot curves (dashed) drop mid-network; shown for completeness only, and we draw no mechanistic conclusion from the few-shot decodability.

scope (C4 .70); the composing and failing models are never in the same format.

F Proof of Proposition 1

The cross-validated coefficient of determination we report is $R^2 = 1 - \mathrm{SS}_{\mathrm{res}}/\mathrm{SS}_{\mathrm{tot}}$ with $\mathrm{SS}_{\mathrm{res}} \geq 0$, so $R^2_{\mathrm{real}} \leq 1$. The baseline probe is a restriction of the same probe class to features F, hence R^2_F is well defined on the same folds. Then gap $= R^2_{\mathrm{real}} - R^2_F \leq 1 - R^2_F$, the headroom. Thus if the baseline already explains the target on D ($R^2_F \to 1$), the gap is forced to 0 regardless of whether the target is separately represented; the test is informative only in proportion to the headroom, which must therefore be reported. At $[20, 49]$ headroom $= 1 - 0.968 = 0.032$ (near-vacuous); at $[2, 99]$ headroom $= 1 - 0.865 = 0.135$, inside which the observed gap 0.060–0.068 (CI excludes 0) lies. □

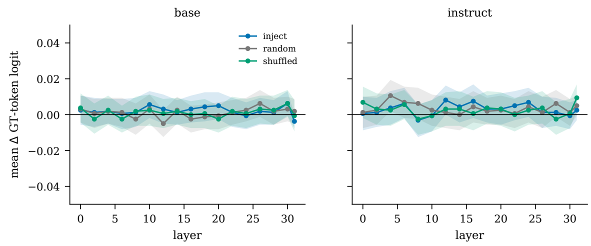

Figure 5: Probe-direction steering at "=" is inert at every layer for inject, random, and shuffled targets (mean Δ ground-truth-token logit, 95% CIs through zero). This first-token metric is a null; the discriminative full-product nulls are the dose-response (§5) and Figure 2b.

G A decodability-audit protocol

The claim "value v is computed at site s" from a decodable probe should clear three checks. (1) **Selectivity with headroom.** Report a gap over a stated baseline feature class and the headroom $1 - R^2_F$ (Prop. 1); a gap in a headroom-less regime is uninformative. (2) **A causal test on a discriminative metric with a same-axis reference that moves.** Steer the decodable direction and score a metric that distinguishes the target from the status quo (here a full-product log-probability, not a shared first token), and show the *same* axis moves a reference where the value is used. (3) **A positive control at a site that moves the answer**, so a null is a dead direction, not a dead instrument. Our answer-site subspace passes (1)'s form but fails (2) and (3) at "=".

H Verbatim prompt strings

Bare-continuation format (`prepend_bos=True`), shown for $B{=}23, C{=}47$; spacing is exact.

```
C0:  23 * 47 =
C1:  ( 0 + 23 ) * 47 =
C2:  ( 0 + 23 * 47 ) =
C3:  ( 23 ) * 47 =
C4:  ( 23 * 47 ) =
C5:  0 + 23 * 47 =
C6:  ( 0 + 23 ) =
C7:  (0+23)*47=
C8:  (23*47)=
A1:  ( 0 + 23 ) + 47 =
A2:  0 + ( 23 + 47 ) =
D1:  (( 0 + 23 )) * 47 =
```

The few-shot C1 template prepends k solved C1 lines (distinct operands, correct answers), newline separated, then the target C1 line; e.g. `( 0 + 20 ) * 21 = 420\n( 0 + 23 ) * 47 =`.

I The answer site is bypassed

After the full-residual swap at the C4 "=" site, the model emits neither the donor product (emit-$P' = 0$ at every layer) nor a corrupted output: it emits its *own* correct answer at 0.90–0.95 across L19–L31 (Table 4). The site is demonstrably read past.

	base		instruct	
layer	logpΔ	emit-true	logpΔ	emit-true
19	$+0.59$	0.90	$+0.12$	0.95
23	$+0.66$	0.92	$+0.15$	0.92
27	$+0.53$	0.92	$+0.13$	0.93
31	$+0.00$	0.92	$+0.00$	0.92

Table 4: Full-residual swap at C4 "=": emit-$P' = 0$ at every layer (omitted); the model keeps emitting its own correct answer. `causal_steering_lateswap_*.json`.

J Baseline scope and reproducibility

Baseline scope. The linear-(B, C) baseline's span excludes B^2, C^2, and $B \cdot C$, so the band-$[2, 99]$ gap upper-bounds operand-*nonlinear* structure and is consistent with, but does not identify, a product component; a quadratic-feature baseline is the decisive test. **Reproducibility.** All results use greedy decoding, integer parsing (parse-fail counts as incorrect), $N = 400$ operand pairs, seed 0, `prepend_bos=True`, `pairs_sha=db4da1`, on Llama-3.1-8B base (`meta-llama/Llama-3.1-8B`) and instruct (rev `0e9e39f`) in bf16 on an NVIDIA A100; TransformerLens 3.5.0, torch 2.11, transformers 5.12.